

# MGT 413 Operations Management

Winter 2026

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## DESCRIPTION

Operations Management (OM) focuses on the systematic design, execution, and continuous improvement of business processes, projects, and partnerships. In today's highly competitive landscape, effective operations management is critical for organizations of all types. This course moves beyond the traditional emphasis on cost minimization to explore the fundamental challenges that organizations must address to achieve scalable, sustainable growth and profitability.

## OBJECTIVES

At the end of this course, you will be able to:

- Understand operations as a process and how it aligns with strategy.
- Apply and interpret key operational metrics in real-world contexts.
- Analyze processes and the relationships among operational metrics.
- Select appropriate methods for evaluating and improving processes.
- Understand the role of inventory and its impact on operations.
- Apply Lean Operations concepts to improve efficiency.
- Recognize how inventory, capacity, and flexibility help mitigate operational risks.

## MATERIALS

- *Managing Business Process Flows: Principles of Operations Management* by Anupindi et al., 3<sup>rd</sup> Edition, Pearson (ISBN: 978-0136036371).
- *Operations Management Course Pack* provided by the Rady School of Management.

## GRADING

| Assessment                       | Points     |
|----------------------------------|------------|
| Class Participation              | 10         |
| Homework Assignments             | 10         |
| Littlefield Technologies         | 15         |
| National Cranberry Case Write-up | 15         |
| Quizzes                          | 20         |
| Final Exam                       | 30         |
| <b>Total</b>                     | <b>100</b> |

According to the *General Student Handbook for the Rady MBA and Specialized Masters Programs*, this course will be graded on a curve. Your final grade will be determined by your performance relative to the rest of the class. The grade distribution for core courses is as follows: 15% A; 20% A–; 30% B+; 20% B; and 15% B–, C+, C, C–, D, or F.

## TENTATIVE SCHEDULE

**CP:** Course Pack; **MBPF:** Managing Business Process Flows. The schedule is subject to change.

| Week                                                   | Due        | Topics and Activities                                    | Assignments and Readings                                                                                                                                   |
|--------------------------------------------------------|------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 1                                                 |            | Introduction<br>Strategic Framework                      | <b>CP:</b><br>- Walmart's Ten-Year Investment in the Frontline<br>- Southwest Airlines: The Next Frontier<br><b>MBPF:</b> §1 & 2                           |
| Week 2                                                 |            | Operations Metrics<br>Capacity Analysis                  | <b>CP:</b> State Automobile License Renewal<br><b>MBPF:</b> §3 & 4                                                                                         |
| Week 3                                                 | HW #1      | Flow-time Analysis<br>Inventory Build-Up                 | <b>CP:</b> National Cranberry Cooperative<br><b>MBPF:</b> §5                                                                                               |
| Week 4                                                 | Project I  | National Cranberry Case<br>Littlefield Technologies (LT) | <b>CP:</b> LT Overview and Capacity Management at LT                                                                                                       |
| Week 5                                                 | HW #2      | Inventory Analysis                                       | <b>CP:</b> Shein Disrupts Fast Fashion and Confronts Sustainability<br><b>MBPF:</b> §6                                                                     |
| Week 6                                                 |            | Inventory pooling<br>Newsvendor                          | <b>CP:</b> DB Sirens Coffee Beans<br><b>MBPF:</b> §7                                                                                                       |
| Week 7                                                 | HW #3      | Queuing and Safety<br>Capacity<br>House Build Game       | <b>CP:</b> Improving Access at VA<br><b>MBPF:</b> §8                                                                                                       |
| Week 8                                                 |            | Beer Game<br>Supply Chain Management                     | <b>CP:</b> The Bullwhip Effect in Supply Chains                                                                                                            |
| Week 9                                                 |            | Lean Operations<br>Sustainability in Operations          | <b>CP:</b> How Circularity Can Be a Strategic Response to Tariffs<br><b>MBPF:</b> §10                                                                      |
| Week 10                                                | Project II | AI in Operations                                         | <b>CP:</b><br>- How 3 Companies Digitized Their Procurement Processes<br>- The Future in Sight: LumineticsCore and the First Autonomous AI for Diagnostics |
| <b>Final Exam: March 17<sup>th</sup>, 6:30–9:30 PM</b> |            |                                                          |                                                                                                                                                            |

## ASSESSMENTS

| Collaboration Policy | Description                                                                                                 |
|----------------------|-------------------------------------------------------------------------------------------------------------|
| C                    | Collaboration with classmates is allowed. However, each student must submit their own individual solutions. |
| G                    | Students may work together in groups and turn in one project or assignment for the entire group.            |
| I                    | Independent, individual work only. No collaboration allowed.                                                |

### Homework Assignments (C)

There will be three homework assignments distributed throughout the quarter. Assignments are due at the start of class on their due date.

### Reading Assignments (C)

No submission of report is required. However, completing assigned readings before class is required to provide essential context. You are expected to prepare responses to guiding questions uploaded to Canvas.

### Team Projects (G)

- **Project I: National Cranberry Case Write-up.** Each team submits a single report. Recommendations should be supported by robust arguments, evidence, and mathematical analysis.
- **Project II: Littlefield Technologies.** A simulation game consisting of two rounds (Capacity Scenario and Forecasting/Inventory Scenario). After the second simulation, each team will submit a written report analyzing their approach.

### Quizzes and Exams (I)

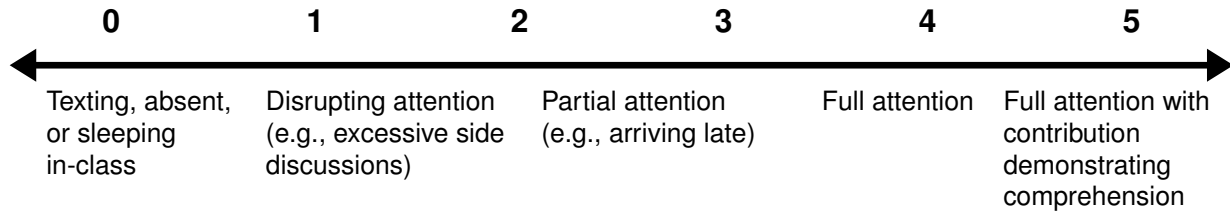
- **Quizzes:** There will be three quizzes throughout the quarter, administered at the beginning of class. Quizzes will be open-note and conducted through Canvas. The lowest quiz score will be dropped.
- **Final Exam:** A three-hour exam. You are allowed to bring one letter-sized sheet of paper with notes on both the front and back.

## COURSE POLICIES

### Participation

Contribution grades are not assigned randomly, nor are they based on one or two memorable class sessions. Rather, they reflect your cumulative track record of preparing for class and actively contributing. This is something I pay close attention to and keep regular notes on. While the

final score will reflect your semester-long performance in this area, the following rubric provides a helpful way to think about contribution on an individual class-by-class basis:



### Electronic Device Policy

You are welcome to use laptops, tablets, or other devices in class to support your learning, such as taking notes or referencing discussion materials. However, using devices for non-class activities (e.g., browsing, messaging) can disrupt the class environment and may negatively impact your grade.

**Phones:** Please keep phones tucked away during class unless you have notified me in advance of an emergency. Unapproved phone use during class will significantly affect your grade.

### AI and ChatGPT Policy

You are welcome to try to use ChatGPT (or similar tools) to offer advice or additional support on your homework and cases. However, you must ultimately do the work on your own. For group homework, the TA will check your writing and raise flags if it is a copy of ChatGPT/AI output. The quiz and final exams are strictly independent work without AI.

### Academic Integrity

Integrity of scholarship is essential for an academic community. All academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind. The complete UCSD Policy on Integrity of Scholarship can be viewed at the Senate Manual.

### Students with Disabilities

A student who has a disability or special need must register with the Office for Students with Disabilities (OSD). The student must present the OSD letter of certification to the faculty member to initiate the request. No accommodations can be implemented retroactively.